
IIO-13180 Systems Simulation

Credits

6.0 (3 hrs lecture)

Lead Instructor

David F. Muñoz

Textbook

Simulation (5th ed.). S. M. Ross. Prentice Hall, 2010.

Supplemental materials

Simio & Simulation: Modeling, Analysis and Applications (3rd ed.). W. D. Kelton, J. S. Smith, D. T. Sturrock, 2014.

Introducción a la Ingeniería (2nd ed.) , S. Romero, O. Romero, D. F. Muñoz. Cengage Learning, 2015

Course Description

Principles and training required to develop a project using computer simulation as an analytical tool.

Fundamentals of discrete event simulation, generation of random variables, input analysis, output analysis and training in the use of high level software to design, implement and validate a simulation model.

Prerequisites

Statistical Inference (EST 11102).

Major Status

Required

Course Outcomes

1. Develop algorithms to generate uniform and non-uniform random-variates.
2. Simulate a non-stationary Poisson process.
3. Build a simulation model using a spreadsheet.
4. Use a software to select an input probability distribution.
5. Compute and interpret a half-width from the output of a terminating simulation.

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7. Perform a test of paired data to compare alternative system designs.
8. Build a simulation model using commercial simulation software.

ABET Outcomes

- a) An ability to apply knowledge of mathematics, science and engineering.

Course Topics

- Review of probability and stochastic processes.
- Building simulation models using a spreadsheet.
- Random number generation.
- Random variate generation.
- Input modeling.
- Output analysis.
- Building simulation models using commercial simulation software.